

MUSTAFA MUDASSER

Python Developer (Backend & Full-Stack | Financial Services | AI/ML)

mustafa.ch.mudasser@gmail.com | 07770 200 851 | Bournemouth, UK (Open to relocation) |
linkedin.com/in/mustafa-mudasser

PROFESSIONAL SUMMARY

Python Software Engineer with 4+ years of production experience building backend systems, data-intensive pipelines, and full-stack applications across the Traditional Banking & Fintech industry in the UK, USA, and MENA. Proven track record delivering Python and React/TypeScript features to Tier 1 investment banking engineering standards at JPMorgan Chase. Experienced with AWS (ECS, EKS, Lambda, EC2, S3), CI/CD, and large-scale data engineering using Python and SQL. Familiar with Apache Spark for data-intensive processing. Currently completing an MSc in Data Science and AI at Bournemouth University (June 2026), with a dissertation on explainable ML for regulated financial services. A fast-moving, collaborative engineer who thrives in cross-functional Agile teams.

CORE COMPETENCIES

+ Python Backend Development	+ React & TypeScript (Full-Stack)	+ Financial Services Engineering
+ AWS (ECS, EKS, Lambda, EC2, S3)	+ Data-Intensive Pipelines	+ Apache Spark (familiar)
+ CI/CD & DevOps (GitHub Actions)	+ RESTful APIs & GraphQL	+ Agile Cross-Functional Teams
+ AI/ML Integration & LLM Systems	+ SQL & Database Optimisation	+ Code Review & TDD/BDD

PROFESSIONAL EXPERIENCE

Software Engineer in Industry | JP Morgan Chase Feb 2026 - May 2026

Tier 1 investment bank. Software Engineering in Industry programme, Bournemouth, UK.

- Delivered production-grade Python/Django backend and React/TypeScript frontend features to JP Morgan Tier 1 investment banking engineering standards within a collaborative, cross-functional Agile team.
- Independently designed Night-Mode Savings from scratch: a Greenfield feature with a 5-minute authorisation window, background worker, and status-polling endpoint, taken from rapid prototype to production in a single sprint.
- Used pytest, GitHub Actions CI/CD, Docker, Git, Postman, and Azure DevOps daily; applied critical thinking across code reviews, branching strategy, and incident diagnosis in staging and production environments.
- Industry research indicated Night-Mode Savings could reduce impulsive purchases by up to 60%; strong stakeholder reception at demo day, with multiple leaders requesting individual follow-up demonstrations.

Python Software Engineer | Grand Digital Pvt Ltd Jan 2025 - Jan 2026

US-based digital agency. Remote contract delivering backend automation and AI-powered systems across multiple business lines.

- Built Python backend automation systems across 4 lines of business, delivering 15+ end-to-end solutions from rapid prototype through to structured platform development.
- Designed and deployed event-driven, data-intensive pipelines and agentic AI workflows using OpenAI API on Azure, achieving a 40% reduction in manual processing and saving 1,200+ hours annually.
- Built KPI dashboards and management reporting systems; authored 40+ technical documents standardising engineering best practices across the team.

Python Software Developer | Visnext Software Solutions Jan 2023 - Nov 2024

B2B and education technology company, Lahore. Production systems serving enterprise and education clients.

- Built and maintained backend Python systems using FastAPI and Django, RESTful and GraphQL APIs, and async processing in production.

- Designed Greenfield automated data ingestion pipeline: 90% reduction in manual processing, 35% faster turnaround, with Redis caching, Elasticsearch search, and optimised PostgreSQL queries.
- Managed AWS deployments (EC2, S3, RDS, Lambda, DynamoDB, SQS) and CI/CD via GitHub Actions, Jenkins, and Terraform; integrated Google, Meta, and third-party APIs.
- Led code reviews, mentored junior engineers, and contributed to architectural decisions within a small, fast-moving team.

Junior Python Developer | Visnext Software Solutions Jul 2022 - Jan 2023

Promoted from junior to mid-level within 6 months.

- Built Django and FastAPI backend features; refactored data pipelines for 35% faster performance; improved test coverage and maintainability.

Junior Software Developer | Airlift Technologies Jul 2021 - Jul 2022

Logistics and operations technology company, Lahore.

- Built Python/SQL automated reporting pipelines, achieving a 40% reduction in manual processing time across large operational datasets.

KEY PROJECTS

Explainable AI for Financial Decision Validation | MSc Dissertation Jan 2026 - Jun 2026

- End-to-end ML pipeline for regulated financial services: Python, PyTorch, SHAP and LIME explainability, EU AI Act compliant, auditable outputs.
- Full R&D lifecycle: hypothesis, experimentation, statistically validated results, MLflow tracking, and written documentation throughout.

Human-Aware Semantic AI System | Greenfield Personal Project 2025

- Built from scratch: FastAPI backend, Pinecone vector DB, Gemini API, semantic retrieval pipeline with embedding generation, similarity search, and LLM reasoning layer.
- Full cycle from rapid prototyping through to working system: async Python, TypeScript/React frontend, REST API end-to-end.

EDUCATION & CERTIFICATIONS

MSc Data Science and Artificial Intelligence | Bournemouth University Jan 2025 - Jun 2026

- Modules: Applied ML, NLP, Reinforcement Learning, Data Engineering, Statistical Modelling, Cloud Computing.

BSc Computer Science | University of Central Punjab 2018 - 2022

- Graduated with Merit. Core study: algorithms, data structures, operating systems, networks.

British Airways Data Science Job Simulation | Forage, Sep 2025 **Fake News Detection with ML** | Coursera, Aug 2025

TOOLS & TECHNOLOGIES

Python, FastAPI, Django, React, TypeScript, SQL, Apache Spark (familiar), PyTorch, MLflow, SHAP, LIME, OpenAI API, Gemini API, Hugging Face | AWS (Lambda, EC2, S3, ECS, EKS, DynamoDB, SQS, API Gateway), Azure (Functions, App Services, OpenAI), Docker, Kubernetes, Terraform | PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis | pytest, TDD/BDD, GitHub Actions, GitLab CI, Jenkins, Azure DevOps, Git